

Xiaonan Fu

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EDUCATION

University of California, San Diego

Sep 2024 - Present

- M.S. in Computer Science

University of California, San Diego

Sep 2021 - June 2024

- B.S. in Cognitive Science (Specialization in Machine Learning and Neural Computation)

TECHNICAL SKILLS

- **Languages:** Python, C/C++, CUDA, TypeScript/JavaScript, Golang, Rust.
- **Deep Learning:** PyTorch, pybind11, Multi-GPU Training.
- **Robotics:** Linux System Admin, Docker, Basic Unreal Engine/Blender.
- **Development:** Full-stack Web Dev, Electron Desktop Apps, Android Native.

RESEARCH INTERESTS

- VLA models, Motion Generation.
- Data-efficient Learning, Meta-learning
- Architecture-level design for training/inference pipelines.

PROJECT EXPERIENCE

Custom Automatic Differentiation Engine & Capsule Network (2025)

- Developed a from-scratch Automatic Differentiation system in Python to understand the underlying mechanics.
- Implemented Capsule Networks using the custom AD engine
- Conducted a systematic performance analysis between the custom implementation and PyTorch, optimizing tensor operations for efficiency.

DPO Fine-tuning for LLM Alignment (2024)

- Finetuned a 124M parameter GPT-2 model using DPO for human preference alignment.
- Leveraged the OpenRLHF framework and LoRA to achieve efficient finetuning.

LLM Distillation & Multi-GPU Optimization (2024)

- Implemented a knowledge distillation pipeline to compress large language models using Attention Map Alignment.
- Optimized the training loop for Multi-GPU environments.

Physics Simulation Engine with Verlet Integration (2022)

- Built a particle physics simulator from scratch using JS and OpenGL/Graphics Libraries.
- Implemented Verlet Integration to simulate gravity and particle motion constraints with high stability.